Respiration in plants

NCERT Nichod : Biology Practice Series NEET 2025

Anmol Sharma • Lesson 12 • Jan 24, 2025

Respiration in Plants

BIOLOGYLIVE

Sigma Institute

Anmol Sharma
NEET 2025

Structure of Atoms

Date: Jan 24, 2025 **Time:** 1hrs **Total Marks:** 50 X 4 = 200

Topics covered

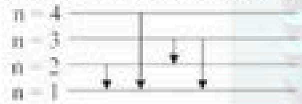
Chemistry

Structure of Atom - Sub-Atomic Particles, Atomic Models, Wave Nature of Electromagnetic Radiation, Particle Nature of Electromagnetic Radiation (EMR), Spectra, Bohr's Model for Hydrogen Atom, Dual Behaviour of Matter - de Broglie Wave Equation, Quantum Numbers, Shapes of Atomic Orbitals, Radial Probability Distribution Curves and Nodes, Filling of Orbitals in Atom, Electronic Configuration of Elements and Magnetic Properties.

Subject : Chemistry **No. of Questions : 50**

1. **Consider the following statements about cathode rays.**
   I. These moves from cathode to anode.
   II. These are visible rays.
   III. Television picture tubes are cathode rays tubes.
   IV. In the absence of electrical or magnetic field, these rays travel in a straight line.
   (1) I, II and III
   (2) II, III and IV
   (3) I, III and IV
   (4) All of these

2. **Suppose that a hypothetical atom gives a red, green, blue and violet line spectrum. Which jump according to the figure would give off the red spectral line?**



   (1) $3 \to 1$
   (2) $2 \to 1$
   (3) $4 \to 1$
   (4) $3 \to 2$

3. **Assertion:** Charge to mass ratio of cathode ray particles is same irrespective of the nature of cathode used or of the gas taken in the discharge tube.
   Reason : Electrons are universal constituents of all matter.
   (1) If both Assertion & Reason are true and reason is the correct explanation of the assertion.
   (2) If both Assertion & Reason are true but reason is not the correct explanation of the assertion.
   (3) If Assertion is true statement but Reason is false.
   (4) Assertion is false but reason is true.

4. **The de-Broglie wavelength associated with a particle of mass 10^{-6}kg with a velocity of $10 ms^{-1}$ is :**
   (1) 6.63×10^{-22}m
   (2) 6.63×10^{-29}m
   (3) 6.63×10^{-31}m
   (4) 6.63×10^{-34}m

5. **The ionisation energy of a hydrogen atom is 13.6 eV. The energy of the third-lowest electronic level in doubly ionised lithium ion ($Z = 3$) is**
   (1) -28.7 eV
   (2) -54.4 eV
   (3) -122.4 eV
   (4) -13.6 eV

6. **Match the orbitals of Column I with the nodal properties of Column II and choose the correct option.**

Column I		Column II	
A	2s	P	Angular node = 2
B	4f	Q	Radial node = 1
C	5d	R	Radial node = 0
D	3p	S	Angular node = 0

   (1) A - S, B - R, C - Q, D - P
   (2) A - S, B - R, C - P, D - Q
   (3) A - R, B - S, C - P, D - Q
   (4) A - R, B - S, C - Q, D - P

7. **According to the de - Broglie equation, the characteristic wavelength of an electron moving around an atomic nucleus will be**
   (1) About the size of the nucleus
   (2) About the same size as the atom
   (3) Large compared to the atomic scale
   (4) Immeasurably small compared to any dimensions we care about

8. **Which of the following options does not represent ground state electronic configuration of an atom?**
   (1) $1s^2 2s^2 2p^6 3s^2 3p^6 3d^8 4s^2$
   (2) $1s^2 2s^2 2p^6 3s^2 3p^6 3d^9 4s^2$
   (3) $1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10} 4s^1$
   (4) $1s^2 2s^2 2p^6 3s^2 3p^6 3d^5 4s^1$

9. **Chlorine exists in two isotopic forms, Cl-37 and Cl-35 but its atomic mass is 35.5. This indicates the ratio of Cl-37 and Cl-35 is approximately**
   (1) 1 : 2
   (2) 1 : 1
   (3) 1 : 3
   (4) 3 : 1

10. **Two atoms are said to be isobars if**
(1) They have same atomic numbers but different mass numbers.
(2) They have same number of electrons but different number of neutrons.
(3) They have same number of neutrons but different number of electrons.
(4) Sum of the number of protons and neutrons is same but the number of protons is different.

11. **Assertion :** The chemical properties of different isotopes are same.
Reason : Isotopes have same number of neutrons.
(1) Both Assertion and Reason are correct and Reason is the correct explanation of the Assertion.
(2) Both Assertion and Reason are correct but Reason is not the correct explanation of the Assertion.
(3) Assertion is correct but Reason is incorrect.
(4) Assertion is incorrect but reason is correct.

12. **A particular station of All India Radio, New Delhi, broadcasts on a frequency of 1,368 kHz (kilohertz). The wavelength of the electromagnetic radiation emitted by the transmitter is :**
[Speed of light, $c = 3.0 \times 10^8 ms^{-1}$]
(1) $0.2192\ m$
(2) $2192\ m$
(3) $21.92\ m$
(4) $219.3\ m$

13. **Assertion :** A spectral line will be observed for a $2p_x - 2p_y$ transition.
Reason : The energy is released in the form of wave of light when electron drops from $2p_x$ to $2p_y$ orbital
(1) Both Assertion and Reason are correct and Reason is the correct explanation of the Assertion.
(2) Both Assertion and Reason are correct but Reason is not the correct explanation of the Assertion.
(3) Assertion is correct but Reason is incorrect.
(4) Assertion is incorrect but Reason is correct.

14. **The energy of an electron in the ground state $(n = 1)$ for He^+ ion is $-xJ$, then that for an electron in $n = 2$ state for Be^{3+} ion in J is**
(1) $-x$
(2) $-\frac{x}{9}$
(3) $-4x$
(4) $-\frac{4}{9}x$

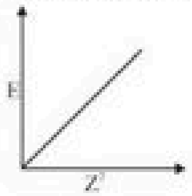
15. **Statement A :** Most of the space in the atoms is empty.
Statement B : Most of the $\alpha-$particles passed through the foil remain undeflected.
(1) Statement A is correct but statement B is incorrect
(2) Statement A is incorrect but statement B is correct
(3) Both statements are correct
(4) Both statements are incorrect

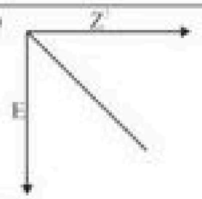
16. **Statement A :** Iron rod is heated in furnace it first turns to dull red, intense red, white and then become blue as the temperature becomes high.
Statement B : The frequency of emitted radiation goes from a higher frequency to a lower frequency as the temperature increases.
(1) Statement A is correct but statement B is incorrect
(2) Statement A is incorrect but statement B is correct
(3) Both statements are correct
(4) Both statements are incorrect

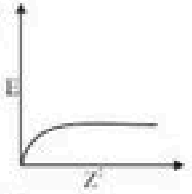
17. **Consider the following statements:**
I. The shape of the orbitals is given by magnetic quantum number.
II. In an atom, all electrons travel with the same velocity.
III. If the value of $l = 0$, the electron distribution is spherical
IV. Angular momentum of 1s, 2s, 3s electrons are equal.
Choose the correct statements and select the correct option.
(1) II and IV
(2) I and III
(3) I and II
(4) III and IV

18. **Statement A :** Greater the energy possessed by the photon, lesser will be transfer of energy to the electron and the kinetic energy of the ejected electron.
Statement B : The kinetic energy of the ejected electron is proportional to the frequency of the electromagnetic radiation.
(1) Statement A is correct but statement B is incorrect
(2) Statement A is incorrect but statement B is correct
(3) Both statements are correct
(4) Both statements are incorrect

19. **Select the correct statement from the following:**
A. Atoms of all elements are composed of two fundamental particles.
B. The mass of the electron is 9.10939×10^{-31} kg
C. All the isotopes of a given elements show same chemical properties.
D. Protons and electrons are collectively known as nucleons.
E. Dalton's atomic theory, regarded the atom as an ultimate particle of matter.
Choose the correct answer from the options given below.
(1) C, D and E only
(2) A and E only
(3) B, C and E only
(4) A, B and C only

20. **The energy of an electron moving in n^{th} Bohr's orbit of an element is given by $E_n = \frac{-13.6}{n^2} Z^2$ eV/atom. The graph of E vs Z^2(keeping 'n' constant) will be**
(1)


(2)


(3)


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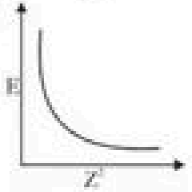
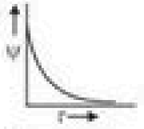
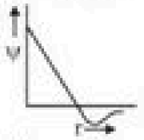


21. **Which of the following graphs correspond to one node ?**

(1)


(2)


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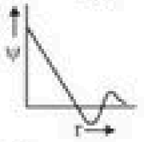
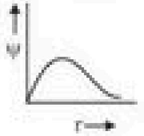
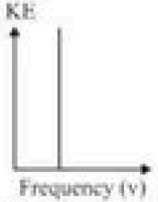
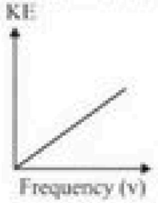
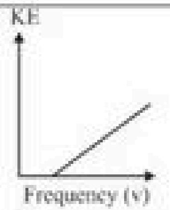


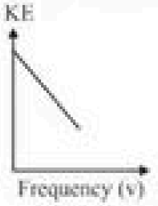
(4)


22. **Which of the following graphs explains photoelectric effect?**

(1)


(2)


(3)


(4)


23. **Match Column I with Column II and choose the correct combination from the options given.**

	Column I		Column II
A	Orbits are filled in order of increasing energy	P	Hund's rule
B	Degenerate orbitals are first singly occupied	Q	Pauli's exclusion principle
C	An orbital can have maximum two electrons	R	Aufbau's principle
D	Position and momentum of a small particle cannot be measured simultaneously with absolute accuracy.	S	Heisenberg's principle

(1) (1) A – R, B – P, C – Q, D – S
(2) (2) A – Q, B – S, C – R, D – P
(3) (3) A – P, B – R, C – Q, D – S
(4) (4) A – S, B – P, C – R, D – Q

24. **Match Column I with Column II and choose the correct combination from the options given.**

	Column I		Column II
A	In the Bohr's hydrogen atom model, the radius of the stationary orbit is directly proportional to (n = principal quantum number)	P	n^2
B	The kinetic energy of the electron in an orbit of radius 'r' in hydrogen atom is (e = electronic charge)	Q	$\frac{e^2}{2r}$
C	The angular momentum of electron in n^{th} orbit is given by	R	$\frac{nh}{2\pi}$
D	According to Bohr's theory, the radius of an orbit described by principal quantum number 'n' and atomic number 'Z' is proportional to	S	$\frac{n^2}{Z}$

(1) (1)A – P, B – R, C – S, D – Q, R
(2) (2) A – P, B – Q, C – R, D – S
(3) (3) A – Q, B – P, C – S, D – R
(4) (4) A – P, B – R, C – S, D – R

25. **Match the column I (element or ion) with column II (It's electronic configuration) and choose the correct option.**

Column I		Column II	
A	Cu	P	$1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10}$
B	Cu^{2+}	Q	$1s^2 2s^2 2p^6 3s^2 3p^6 3d^3$
C	Zn^{2+}	R	$1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10} 4s^1$
D	Cr^{3+}	S	$1s^2 2s^2 2p^6 3s^2 3p^6 3d^9$

(1) A - S, B - R, C - P, D - Q
(2) A - R, B - S, C - P, D - Q
(3) A - Q, B - S, C - R, D - P
(4) A - R, B - Q, C - P, D - S

26. **Match the type of series given in Column I with the wavelength range given in Column II and choose the correct option**

	Column I		Column II
A	Lyman	P	Ultraviolet
B	Paschen	Q	Near infrared
C	Balmer	R	Far infrared
D	Pfund	S	Visible

(1) A – P, B – Q, C – S, D – R
(2) A – S, B – R, C – P, D – Q
(3) A – R, B – P, C – Q, D – S
(4) A – S, B – R, C – Q, D – P

27. **Match Column I (Alkali Metal) with Column II (Emission Wavelength in nm).**

Column I		Column II	
A	Li	P	589.2
B	Na	Q	455.5
C	Rb	R	670.8
D	Cs	S	780.0

(1) A – S, B – Q, C – P, D – R
(2) A – Q, B – S, C – R, D – P
(3) A – R, B – P, C – S, D – Q
(4) A – P, B – S, C – R, D – Q

28. **Match the following**

Column I		Column II	
A	Energy of ground state of He^+	P	+6.04 eV
B	Potential energy of I^{st} atom orbit of H-atom	Q	27.2 eV
C	Kinetic energy of 2^{nd} excited state of He^+	R	54.4 eV
D	Ionisation of potential of He^+	S	-54.4 eV

(1) A – P, B – Q, C – R, D – S
(2) A – S, B – R, C – Q, D – P
(3) A – S, B – Q, C – P, D – R
(4) A – Q, B – R, C – P, D – S

29. **Match the following pairs in the given compound in Column I as ("Atom/ion") and Column II as ("No. of electrons").**

Column I		Column II	
A	O^{2-}	P	18
B	Li^{2+}	Q	2
C	He	R	10
D	Ca^{++}	S	1

(1) A – P, B – Q, C – S, D – R
(2) A – Q, B – R, C – P, D – S
(3) A – R, B – S, C – Q, D – P
(4) A – R, B – P, C – R, D – Q

30. **Match Column I (Quantum number) with Column II (Information provided)**

Column I		Column II	
A	m_l	p	Shape of orbital
B	m_s	Q	Size of orbital
C	l	R	Orientation of orbital
d	n	S	Orientation of spin of electron

(1) A – P, B – R, C – Q, D – S
(2) A – R, B – S, C – P, D – Q
(3) A – R, B – S, C – Q, D – P
(4) A – Q, B – P, C – S, D – R

31. **Match Column I (Spectral Series for Hydrogen) with Column II (Spectral Region/Higher Energy State).**

Column I		Column II	
A	Lyman	P	Infrared region
B	Balmer	Q	UV region
C	Paschen	R	Infrared region
D	Pfund	S	Visible region

(1) A – Q, B – R, C – P, D – S
(2) A – P, B – Q, C – R, D – S

(3) $A-P, B-R, C-Q, D-S$

(4) $A-Q, B-S, C-R, D-P$

32. **The maximum number of electrons with clockwise spin that can be accommodated in a f-subshell is**

(1) 14

(2) 7

(3) 5

(4) 10

33. **If the shortest wavelength in Lyman series of H atom is x, then longest wavelength in Balmer series of He^+ is**

(1) $\frac{9x}{5}$

(2) $\frac{36x}{5}$

(3) $\frac{x}{4}$

(4) $\frac{5x}{9}$

34. **What will be the wave number of yellow radiation having wavelength 240 nm ?**

(1) $1.724 \times 10^4\ cm^{-1}$

(2) $4.16 \times 10^6\ m^{-1}$

(3) $4 \times 10^{14}\ Hz$

(4) $219.3 \times 10^3\ cm^{-1}$

35. **The number of d - electrons in Ni is equal to that of the**

(1) s and p - electrons in F^-

(2) p - electrons in Ar

(3) d - electrons in Ni^{2+}

(4) total number of electrons in N

36. **If the wavelength of photon is 2.2×10^{-11} m $h = 6.6 \times 10^{-34}$ Js, then momentum of photon is**

(1) $3 \times 10^{-23}\ kg\ ms^{-1}$

(2) $3.33 \times 10^{22}\ kg\ ms^{-1}$

(3) $1.452 \times 10^{-44}\ kg\ ms^{-1}$

(4) $6.89 \times 10^{43}\ kg\ ms^{-1}$

37. **A metal surface is exposed to solar radiations**

(1) The emitted electrons has energy less than a maximum value of energy depending upon the frequency of the incident radiation.

(2) The emitted electrons have energy less than the maximum value of energy depending upon intensity of incident radiations.

(3) The emitted electrons have zero energy.

(4) The emitted electrons have energy equal to energy of photons of incident light.

38. **The orbital angular momentum of an electron in the 3d subshell is**

(1) 1.1h

(2) 0.21h

(3) 0.39h

(4) 0.46h

39. **The line spectra of two elements are not identical because**

(1) The elements do not have the same number of neutrons.

(2) They have different mass number.

(3) Their outermost electrons are at different energy levels.

(4) They have different valencies.

40. **An isotone of $^{76}_{32}Ge$ is**

(i) $^{77}_{32}Ge$ (ii) $^{77}_{33}As$

(iii) $^{77}_{34}Se$ (iv) $^{78}_{34}Se$

(1) Only (i) and (ii)

(2) Only (ii) and (iii)

(3) Only (ii) and (iv)

(4) Only (ii), (iii) and (iv)

41. **If the kinetic energy of an electron is increased by 4 times, the wavelength of the de Broglie wave associated with it would become**

(1) 4 times

(2) 2 times

(3) $\frac{1}{2}$ times

(4) $\frac{1}{4}$ times

42. **Wavelength of the first line of Paschen series hydrogen spectrum is ($R = 109700 cm^{-1}$)**

(1) 18750Å

(2) 2854Å

(3) 3452Å

(4) 6243Å

43. **Photons of frequency 3.2×10^{16} Hz is used to irradiate a metal surface, the maximum kinetic energy of the emitted photo-electron is $\frac{3^{th}}{4}$ of the energy of the irradiating photon. What is the threshold frequency of the metal?**

(1) $2.4 \times 10^{25}\ Hz$

(2) $2.4 \times 10^{16}\ Hz$

(3) $1.6 \times 10^{15}\ Hz$

(4) $8 \times 10^{15}\ Hz$

44. **Monochromatic radiation of specific wavelength is incident on H-atoms in ground state. H-atoms absorb energy and emit subsequently radiations of six different wavelength. Find wavelength of incident radiations:**

(1) 9.75 nm

(2) 50 nm

(3) 85.8 nm

(4) 97.25 nm

45. **Electronic transition in Li^{2+} ion takes from n_2 to n_1 shell when electrons transition to 1^{st} shell from 3^{rd} shell, the number of spectral line formed are**

(1) 10

(2) 8

(3) 5

(4) 3

46. **A gas absorbs photon of 355 nm and emits at two wavelengths. If one of the emission is at 680 nm, the other is at**

(1) 1035 nm

(2) 325 nm

(3) 743 nm
(4) 518 nm

47. The radius of the first orbit of hydrogen atom is 0.52×10^{-8} cm. The radius of the first orbit of He^+ ion is

(1) 0.26×10^{-8} cm
(2) 0.52×10^{-8} cm
(3) 1.04×10^{-8} cm
(4) 2.08×10^{-8} cm

48. Which one is a wrong statement?

(1) $1s^2$ [↑↓] $2s^2$ [↑↓] $2p_x^1\ 2p_y^1\ 2p_z^1$ [↑ | ↑ | ↓]
The electronic configuration of N atom is
(2) The value of m for $d_z{}^2$ is zero
(3) An orbital is designated by three quantum numbers while an electron in an atom is designated by four quantum numbers.
(4) Total orbital angular momentum of electron in 's' orbital is equal to zero.

49. The quantum number ' m ' of a free gaseous atom is associated with

(1) The effective volume of the orbital
(2) The shape of the orbital
(3) The spatial orientation of the orbital
(4) The energy of the orbital in the absence of a magnetic field

50. A bulb emits electromagnetic radiation of 660 nm wavelength. The total energy of radiation is 3×10^{-18} J. The number of emitted photons will be $(h = 6.6 \times 10^{-34}\ J \times s, c = 3 \times 10^8\ m/s)$

(1) 1
(2) 10
(3) 100
(4) 1000

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1. Given below are two statements :

Statement I :

In respiration energy of oxidation reduction utilised for phosphorylation

Statement II :

Number of ATP synthesised depends on nature of electron donor

Choose the correct answer from the options given below.

(1) Both Statement I and Statement II are incorrect

(2) Statement I is correct but Statement II isincorrect

(3) Statement I is incorrect but Statement II is correct

(4) Both, Statement I and Statement II are correct

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2. Given below are two statements :

Statement I : Organism like some bacteria pyruvic acid is convert to CO_2 and lactic acid

Statement II : ATP is not synthesised during the conversion of PEP to puruvic acid

Choose the correct answer from the options given below.

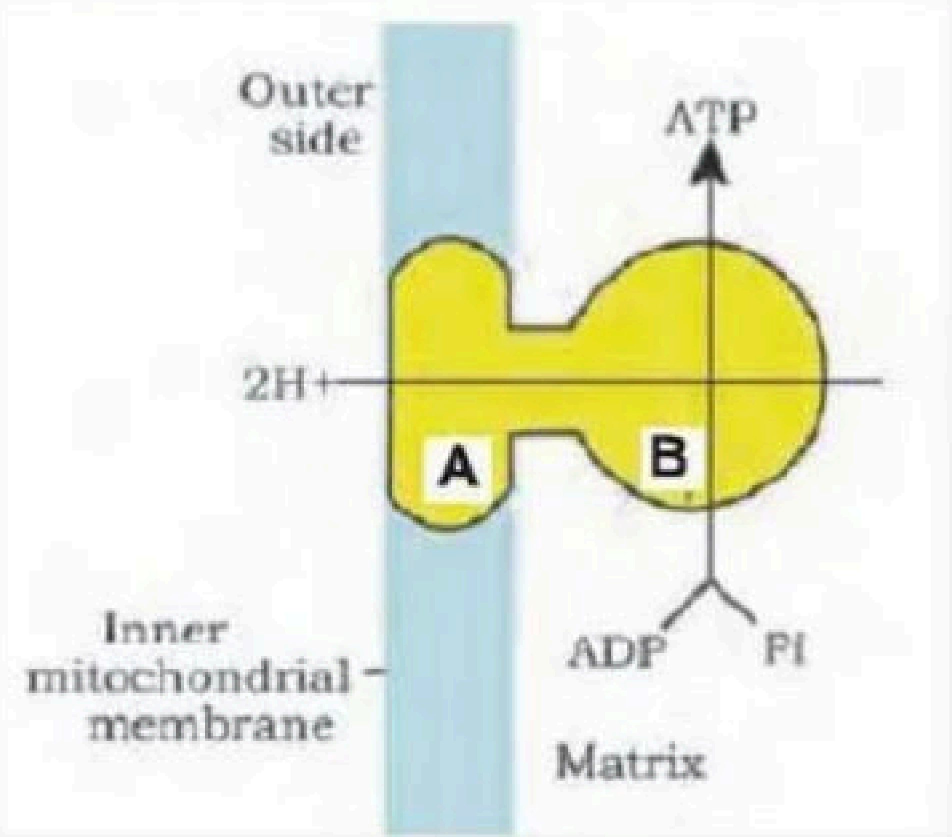
(1) Both Statement I and Statement II are incorrect

(2) Statement I is correct but Statement II isincorrect

(3) Statement I is incorrect but Statement II is correct

(4) Both, Statement I and Statement II are correct

3. In given figure A and B are component of which complex :



(1) Complex I (2) Complex III

(3) Complex V (4) Complex II

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4. Which one of the following process releases a carbon dioxide molecule:

(1) Glycolysis

(2) Lactic acid fermentation

(3) Alcoholic fermentation

(4) Both 2 and 3

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5. How many NADH + H^+ will be produced during the production of molecule of acetyl CoA from two molecule of pyruvic acid :

(1) 1 NADH + H^+

(2) 2 NADH + H^+

(3) 3 NADH + H^+

(4) 4 NADH + H^+

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6. In ETS, complex-V is :

(1) NADH dehydrogenase

(2) ATP synthase

(3) Succinate dehydrogenase

(4) Ubiquinone

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7. Given below are two statements :

Statement I : Ubiquinol is oxidised with the transfer of electrons to cytochrom c via cytochrome bc1 complex

Statement-II : Cytochrome c oxidase complex contain cytochromes a and a3 and two copper centres.

Choose the correct answer from the option given below:

(1) Both Statement I and Statement II are incorrect

(2) Statement I is correct but Statement II is incorrect

(3) Statement I is incorrect but Statement II is correct

(4) Both Statement I and Statement II are correct

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8. Which one of the following is not included in glycolysis:

(1) Substrate level phosphorylation occurs

(2) The end products are CO2 and H2O.

(3) ATP is formed.

(4) ATP is used.

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9. Out of 38 molecules of ATP produced during aerobic respiration of glucose, the net ATP production (directly or indirectly) in glycolysis (A), pyruvate to acetyl Co-A formation (B) and Kreb's cycle (C) is :

(1) A = 1, B = 6, C = 30

(2) A = 8, B = 6, C = 24

(3) A = 8, B =10, C = 20

(4) A = 2, B =12, C = 24

Title: Respiration in Plants

10. Which of the following compounds enter in the glycolytic pathway:

(1) Glucose

(2) Fructose

(3) Sucrose

(4) Both 1 and 2

Title: Respiration in Plants

11. Starting from sucrose upto the production of glucose-6-phosphate, which enzyme(s) is/are involved:

(a) Invertase

(b) Hexokinase

(c) Isomerase

(d) Kinase

(1) a, b (2) a, c

(3) b, c (4) a, b, c

Title: Respiration in Plants

12. The process by which partial oxidation of glucose occurs:

(1) Kreb's cycle

(2) E.T.S

(3) Both (1) and (2)

(4) Glycolysis

Title: Respiration in Plants

13. Enzymes which catalyze the conversion of pyruvic acid into CO_2 and ethanol in fermentation are

(1) Pyruvic acid decarboxylase

(2) Alcohol dehydrogenase

(3) Both (1) and (2)

(4) Hexokinase

Title: Respiration in Plants

14. Substrate level phosphorylation occurs in glycolysis when there is conversion of

(1) BPGA → PGA and phosphoglycerate → PEP

(2) BPGA → PGA and PEP → Pyruvic acid

(3) Fructose-6-phosphate → Fructose 1,6-biphosphate

(4) Glucose → Glucose-6-phosphate

Title: Respiration in Plants

15. At how many places in the kreb cycle is NAD^+ reduced to $NADH+H^+$, and FAD^+ reduced to $FADH_2$, respectively:

(1) One and three

(2) Three and one

(3) Four and one

(4) Three and two

Title: Respiration in Plants

16. What happens when succinyl-CoA is converted into succinic acid

(1) Reduction of NAD^+ to $NADH + H^+$

(2) Conversion of $NADH + H^+$ to NAD^+

(3) Conversion of FAD^+ to $FADH_2$

(4) Conversion of GDP+IP to GTP

Title: Respiration in Plants

17. Total indirect ATP synthesized in glycolysis :

(1) 2

(2) 4

(3) 6

(4) 8

Title: Respiration in Plants

18. During which stage in the complete oxidation of glucose are the greatest number of ATP molecules formed from ADP

(1) Conversion of pyruric acid to acetyl Co A

(2) Electron transport chain

(3) Glycolysis

(4) Krebs cycle

Title: Respiration in Plants

19. In which one of the following do the two names refer to one and the same thing :

(1) Kreb's cycle and Calvin cycle

(2) Tricarboxylic acid cycle and citric acid cycle

(3) Citric acid cycle and Calvin cycle

(4) Tricarboxylic acid cycle and urea cycle

20. R.Q. of maturing fatty seeds will be

(1) 1

(2) More than one

(3) 0

(4) 0.7

Title: Respiration in Plants

21. How many ATP molecules produced by aerobic oxidation of one molecule of glucose :

(1) 2

(2) 4

(3) 38

(4) 34

Title: Respiration in Plants

22. Direct gain of ATP from one mole of glucose during glycolysis or EMP pathaway

(1) 2 ATP

(2) 6 ATP

(3) 36 ATP

(4) 38 ATP

Title: Respiration in Plants

23. In kreb's cycle two successive steps of decarboxylation, leading to the formation of ultimately

(1) -ketoglutaric acid

(2) Citrate

(3) Isocitrate

(4) Succinyl-CoA

Title: Respiration in Plants

24. Substrate level phosphorylation in Kreb's cycle takes place in between

(1) Citrate to isocitrate.

(2) α -ketoglutaric acid to succinyl-CoA

(3) Conversion of succinyl-CoA to succinic acid

(4) Succinic acid to fumaric acid

25. RQ of this reaction

$2(C_{51}H_{98}O_6) + 145O_2 \rightarrow 102CO_2 + 98H_2O$ + Energy,

would be

(1) One

(2) Less then one

(3) Infinite

(4) More then one

Title: Respiration in Plants

26. Electrons from NADH produced in the mitochondrial matrix during citrtc acid cycle are oxidised by

(1) NADH dehydrogenase (complex I)

(2) $FADH_2$ (complex II)

(3) Ubiquinone (ubiquinol)

(4) Cytochrome bc_1 complex (complex III)

27. Cytochrome c is found between, in ETS

(1) Complex I and II

(2) Complex II and III

(3) Complex III and IV.

(4) Complex IV and V

Title: Respiration in Plants

28. Given below are two statements : one is labelled as Assertion (A) and the other is labelled as Reason (R)

Assertion (A) : Respiratory pathway is considered as an amphibolic pathway

Reason (R) : It involves both anabolism and catabolism.

choose the correct answer from the options given below :

(1) Both (A) and (R) are correct but (R) is not the correct explanation of (A)

(2) (A) is correct but (R) is not correct

(3) (A) is not correct but (R) is correct

(4) Both (A) and (R) are correct and (R) is the correct explanation of (A)

Title: Respiration in Plants

29. Which is the connecting link between glycolysis and Kreb's cycle :

(1) Iso-citric acid

(2) α-ketoglutaric acid

(3) Glucose

(4) Acetyl Co A.

Title: Respiration in Plants

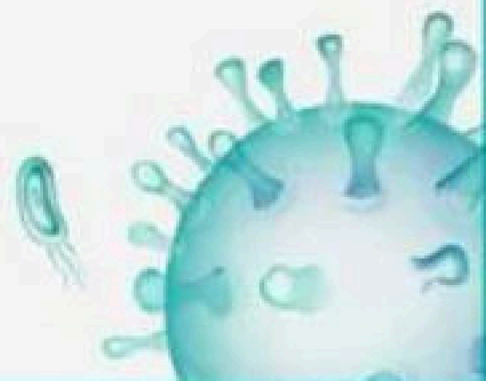
30. Raw material for respiration is :

(1) Glucose and CO_2

(2) Glucose and O_2

(3) Glucose and carbon

(4) Glucose and sucrose



31. Match the column I and column II :

Column I	Column II
a. 2C compound	i. Citric acid
b. 3C compound	ii. Pyrubic acid
c. 4C compound	iii. Malic acid
d. 5C compound	iv. Acetyl CoA
	v. – Ketoglutaric acid

(1) a – iv, b – iii, c – ii, d – v
(2) a – iv, b – ii, c – iii, d – i
(3) a – iv, b – ii, c – iii, d – v
(4) a – ii, b – iii, c – iv, d – i

Title: Respiration in Plants

32. Number of CO_2 molecules evolved in glycolysis is:

(1) 2

(2) 1

(3) 3

(4) 0

Title: Respiration in Plants

33. Sucrose is converted into glucose and fructose by the enzyme:

(1) Invertase

(2) Zymage

(3) Hexokinase

(4) Ligase.

Title: Respiration in Plants

34. Which of these statements is incorrect :

(1) Enzyme of TCA cycle are present in mitochondrial matrix

(2) TCA cycle occurs in mitochondrial matrix

(3) Glycolysis occurs in cytoplasm

(4) Oxidative phosphorylation takes place in outer membrane of mitochondria

Title: Respiration in Plants

35. How many pyruvic acid will formed from 1 mole of 3-phosphoglyceric acid during glycolysis :

(1) 1

(2) 2

(3) 3

(4) 4

Title: Respiration in Plants

36. Cytochrome c is a small protein attached to the:

(1) Outer surface of the inner membrane

(2) Inner surface of the outer membrane

(3) Inner surface of the inner membrane

(4) Outer surface of the outer membrane

Title: Respiration in Plants

37. In any case (aerobic or anaerobic), all living organisms retain the enzymatic machinery to partially oxidize glucose without the help of oxygen. This breakdown of glucose is called:

(1) Glycogenesis

(2) Glycogenolysis

(3) Glycolysis

(4) Gluconeogenesis

Title: Respiration in Plants

38. For two molecules of glucose, glycolysis uses and produces how many ATP molecules?

(1) 4 and 8

(2) 2 and 4

(3) 2 and 8

(4) 2 and 2

Title: Respiration in Plants

39. In alcoholic fermentation, by yeast, pyruvic acid is converted to ethyl alcohol and carbon dioxide with the help of the enzyme:

(1) Pyruvate dehydrogenase

(2) Alcohol dehydrogenase

(3) Both (1) and (2)

(4) Pyruvic acid decarboxylase and alcohol dehydrogenase

Title: Respiration in Plants

40. Ethanol is formed from acetaldehyde by an enzyme called:

(1) Lactase dehydrogenase

(2) Pyruvate kinase

(3) Alcohol dehydrogenase

(4) Pyruvate decarboxylase

Title: Respiration in Plants

41. End product(s) of fermentation of sugars is/are:

(1) Water and carbon dioxide

(2) Alcohol and carbon dioxide

(3) Carbon dioxide only

(4) Alcohol only

Title: Respiration in Plants

42. The ratio between net gain of ATP, number of CO_2 liberated, and number of ATP utilized during anaerobic respiration with alcohol fermentation is:

(1) 1:2:3

(2) 2:1:2

(3) 1:1:1

(4) 1:2:1

Title: Respiration in Plants

43. Proteins act as respiratory substrates and get degraded by proteases into individual amino acids (after deamination). Depending on their structure, they would enter the pathway at some stage:

(1) As acetyl CoA

(2) As pyruvate

(3) Within the Krebs' cycle

(4) Any of the above

Title: Respiration in Plants

44. Number of glucose molecules required to produce 38 ATP molecules under anaerobic conditions is:

(1) 38

(2) 4

(3) 19

(4) 2

Title: Respiration in Plants

45. R.Q. values of 4, 1, and 0.7 occur in the case of:

(1) Malic acid, palmitic acid, and tripalmitin

(2) Oxalic acid, carbohydrate, and tripalmitin

(3) Tripalmitin, malic acid, and carbohydrate

(4) Palmitic acid, carbohydrate, and oxalic acid

Title: Respiration in Plants

46. R.Q. is maximum when the respiratory substrate is: (1) Glucose

(2) Fat

(3) Protein

(4) Malic acid

Title: Respiration in Plants

47. What are the reasons why plants can get along without respiratory organs?

A. Each plant part takes care of its own gas exchange needs. There is very little transport of gases from one plant part to another.

B. Plants do not present great demands for gaseous exchange: root, stem, and leaves respire at rates far lower than animals do.

C. The distance that gases must diffuse even in large, bulky plants is not great.

(1) A and B

(2) B and C

(3) C and A

(4) A, B, and C

48. There are three major ways in which different cells handle pyruvic acid produced by glycolysis. These are:

(1) Fermentation, TCA, and ETS

(2) Fermentation, aerobic respiration, and TCA

(3) Alcoholic fermentation, lactic acid fermentation, and aerobic respiration

(4) Alcoholic fermentation, lactic acid fermentation, and ETS

Title: Respiration in Plants

49.In animal cells, like muscles during exercise, when oxygen is inadequate for cellular respiration, pyruvic acid is reduced to lactic acid by:

(1) Lactate decarboxylase

(2) Pyruvate dehydrogenase

(3) Both (1) and (2)

(4) Lactate dehydrogenase

50. Fermentation is essentially glycolysis plus an extra step in which pyruvic acid is reduced to form lactic acid or alcohol and CO_2. This last step:

(1) Removes poisonous oxygen from the environment.

(2) Extracts a bit more energy from glucose.

(3) Enables the cell to recycle NAD^+.

(4) Inactivates toxic pyruvic acid.

Title: Respiration in Plants

51. In yeast cell continious fermentation of glucose causes death of yeast cell :

(1) Suffocation

(2) Unavailability of oxygen

(3) High ethyl alcohol

(4) All of these

Title: Respiration in Plants

52. Choose the correct combination between respirotory substrate and their respective RQs:

	Fat	Carbohydrate	Protein
(1)	1	0.7	0.9
(2)	0.7	1	0.9
(3)	0	1	0.7
(4)	0.7	2	0.9

Title: Respiration in Plants

53. Match the following :

a. Cyt bc_1	P. Complex V
b. ATP synthase	Q. Complex I
c. Cyt c oxidase	R. Complex III
d. NADH – Dehydrogenase	S. Comples IV

(1) a – R , b – P , c – S, d – Q

(2) a – R , b – S , c – P, d – Q

(3) a – S , b – R , c – Q, d – P

(4) a – R , b – P , c – Q, d – S

Title: Respiration in Plants

54. Which of the following process is used in conversion of pyruvate to acetyl CoA :

(1) Oxidative decarboxylation

(2) Oxidative dehydrogenation

(3) Oxidative dehydration

(4) Oxidative phosphorylation.

Title: Respiration in Plants

55. Which of the following biomolecules is common to respiration mediated breakdown of fatty acid carbohydrates and proteins :

(1) Pyruvic acid

(2) Acetyl CoA

(3) Glyceraldehyde 3 – Phasphate

(4) 1 and 2 both

Title: Respiration in Plants

56. $\frac{\text{Volume of } CO_2 \text{ evolved}}{\text{Volume of } O_2 \text{ consumed}}$

Above ratio is known as :

(1) Respiratory rate

(2) Respiratory quotient

(3) Transpiration quotient

(4) All of the above

Title: Respiration in Plants

57. The overall goal of glycolysis, Krebs cycle and the electron transport system is the formation of:

(1) ATP in small stepwise units

(2) ATP in one large oxidation reaction

(3) Sugars

(4) Nucleic acids

Title: Respiration in Plants

58. Given below are two statements :

Statement I : During oxidative decarboxylation of one molecule of pyruvic acid in mitochondrial matrix one molecule of NADPH + H^+ is produced.

Statement II : In glycolysis, glucose undergoes partial oxidation to form two molecule of pyruvic acid.

Choose the correct answer from the options given below.

(1) Both Statement I and Statement II are incorrect

(2) Statement I is correct but Statement II isincorrect

(3) Statement I is incorrect but Statement II is correct

(4) Both, Statement I and Statement II are correct

Title: Respiration in Plants

59. How many ATP molecules can be produced through oxidative phosphorylation of $2NADH_2$ and $3\ FADH_2$

(1) 15

(2) 24

(3) 6

(4) 12

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Title: Respiration in Plants

60. Which of the following is 5C compound of Kreb's cycle:

(1) α-ketoglutaric acid

(2) Isocitric acid

(3) Cis aconitic acid

(4) Pyruvic acid

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61. Select the wrong statement

(1) When tripalmitin is used as a substrate in respiration, the RQ is 0.7

(2) One glucose molecule yields a net gain of 2 ATP molecules during fermentation

(3) The intermediate compound which links glycolysis with Kreb's cycle is malic acid

(4) One glucose molecule yields a net gain of 38 ATP molecules during aerobic respiration

Title: Respiration in Plants

62. Cytochrome C oxidase complex contains :

(1) Cytochrome a, a_3 and one copper centre

(2) Cytochrome a and two copper centres

(3) Cytochrome a_3 and one copper centre

(4) Cytochrome a, a_3 and two copper centre

Title: Respiration in Plants

63. How many ATP will be produced during the production of molecule of acetyl CoA from two molecule of pyruvic acid :

(1) 3 ATP

(2) 6 ATP

(3) 8 ATP

(4) 38 ATP.

Title: Respiration in Plants

64. Which of the following is incorrectly matched :

(1) Protein → Degarded by proteases

(2) Fats → Fatty acid + Glycerol

(3) Kreb's cycle → Carboxylation

(4) Respiratory → pathway Amphibolic

Title: Respiration in Plants

65. Which is also formed alongwith ATP in glycolysis:

(1) $NADPH_2$

(2) $NADH_2$

(3) $FADH_2$

(4) FAD

66. During the degradation of six glucose molecules in fermentation, the net gain of ATP per glucose molecule degraded is:

(1) 12

(2) 6

(3) 2

(4) 8

Title: Respiration in Plants

67. Which one of the following molecule are utilised as well as released during glycolysis process :

(1) ATP, H_2O

(2) ATP

(3) ATP, H_2O, CO_2

(4) ATP, H_2O, CO_2, O_2

68. All of the following processes can release CO_2 except

(1) Alcoholic fermentation

(2) Oxidative decarboxylation and Krebs' cycle

(3) Oxidative phosphorylation

(4) Conversion of α −ketoglutaric acid to succinic acid

Title: Respiration in Plants

69. Protein enters the Kreb's cycle as

(1) Amino acid after deamination

(2) Pyruvate

(3) Acetyl CoA

(4) All of these

Title: Respiration in Plants

70. When fats are broken down what will formed :

(1) Fatty acid and glycerol

(2) Fatty acid and ethanol

(3) Fatty acid and carbinol

(4) Glycerol and pyruvic acid.

Title: Respiration in Plants

71. The correct sequence of electron acceptor in ATP synthesis is :

(1) Cyt, b, c, a_3, a

(2) Cyt, c, b, a, a_3

(3) Cyt. a, a, b, c

(4) Cyt. b, c_1,c, a, a_3

Title: Respiration in Plants

72. Pyruvic acid is the end product of which process

(1) Kreb's cycle

(2) Calvin cycle

(3) Pentose phosphate pathway

(4) Glycolysis

Title: Respiration in Plants

73. The term glycolysis has originated from the Greek words,

(1) Glucose

(2) Glycos

(3) Glycon

(4) Glycogen

Title: Respiration in Plants

74. Glycolysis, a chain of ..... reactions, under the control of different enzymes

(1) Nine

(2) Ten

(3) Eleven

(4) Twelve

Title: Respiration in Plants

75. The respiratory quotient depends upon

(1) Kind of organism, used in test of respiration

(2) Respiratory pathway, as glycolysis, kreb's cycle and ETS

(3) Type of respiratory substance used during respiration.

(4) Number of oxygen molecules used during reduction of respiratory substrates.

Title: Respiration in Plants

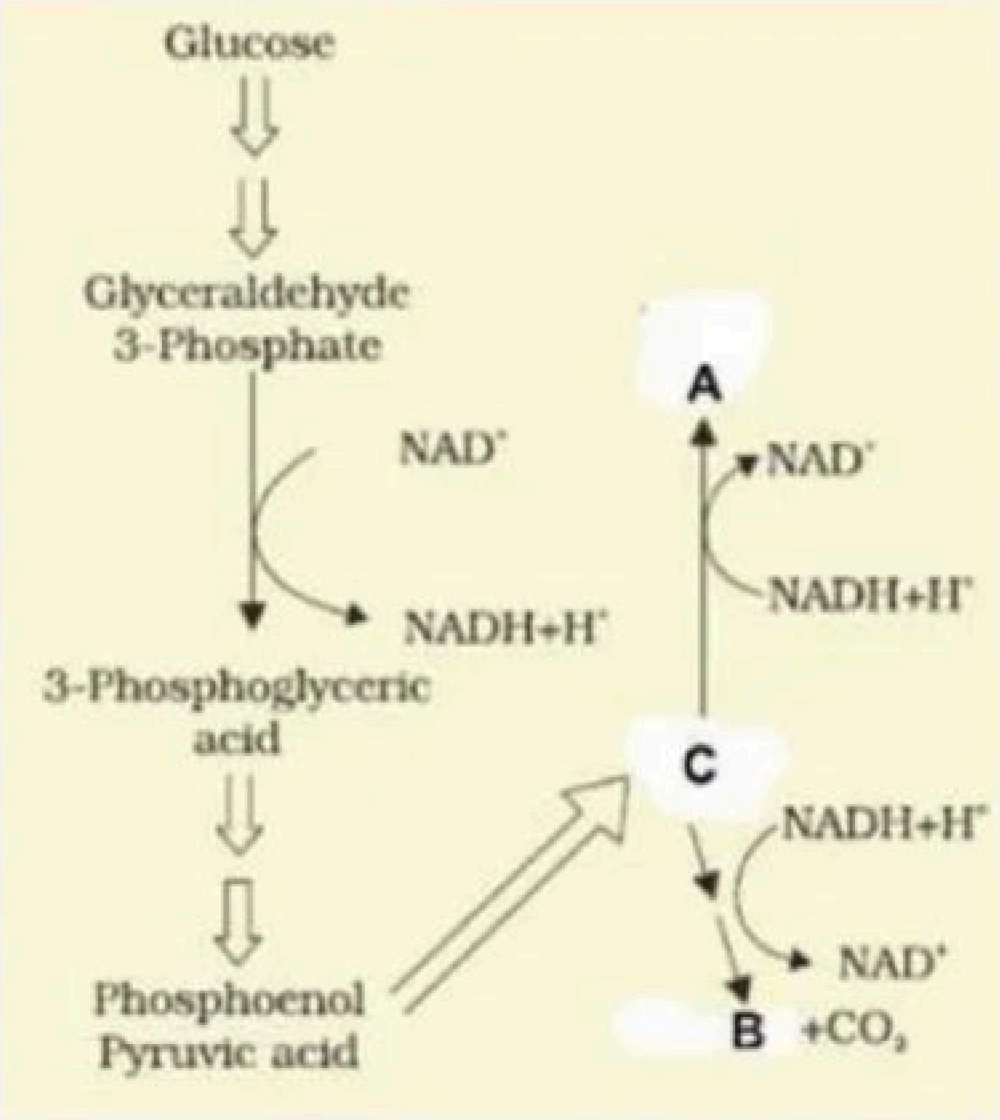
76. In given diagram A and B are :

(1) Lactic acid and ethanol respectively

(2) Ethanol and lactic acid respectively

(3) Pyruvic acid and Ethanol respectively

(4) Lactic acid pyrubic acid respectively



Title: Respiration in Plants

77. In a cell in aerobic respiration what is the correct sequence of different process :

(1) Glycolysis → Oxidative decarboxylation → Kreb's Cycle → ETS

(2) Glycolysis → Oxidative decarboxylation → Kreb's Cycle → TCA cycle →Fermentation

(3) Glycolysis → Fermentation → TCA cycle → ETS

(4) Glycolysis → Fermentation → Kreb's Cycle → ETS

Title: Respiration in Plants

78. The pathway of respiration common in all living organisms is X ; it occurs in the Y and the products formed are two molecules of Z .

Identify X, Y and Z in the above paragraph and select the correct answer.

	X	Y	Z
(1)	Glycolysis	mitochondrion	pyruvic acid
(2)	Glycolysis	cytoplasm	pyruvic acid
(3)	Krebs' cycle	cytoplasm	acetyl CoA
(4)	Krebs' cycle	mitochondrion	acetyl CoA

79. Select the wrong statement with respect to glycolysis.

(1) It occurs outside mitochondria

(2) It is an anaerobic phase

(3) Glucose undergoes partial oxidation to form 2 molecules of pyruvic acid

(4) Glucose is phosphorylated to glucose-6-phosphate by isomerase enzyme

Title: Respiration in Plants

80. Identify A and B in the given reaction.

$$\text{Pyruvic acid} + \text{CoA} + \text{NAD}^+ \xrightarrow[\text{Pyruvate dehydrogenase}]{\text{Mg}^{2+}} \text{A} + \text{B} + \text{NADH} + \text{H}^+$$

A	B
(1) PEP	CO_2
(2) Acetyl CoA	CO_2
(3) CO_2	H_2O
(4) Acetyl CoA	H_2O

Title: Respiration in Plants

81. Match column I with column II and select the correct option from the given codes

Column I	Column II
A. Glycolysis	(i) Inner mitochondrial membrane
B. TCA cycle	(ii) Mitochondrial matrix
C. ETS	(iii) Cytoplasm

(1) A-(iii), B-(i), C-(ii)

(2) A-(iii), B-(ii), C-(i)

(3) A-(i), B-(ii), C-(iii)

(4) A-(ii), B-(i), C-(iii)

Title: Respiration in Plants

82. NADH+ H^+ is produced during glycolysis during conversion of

(1) 3-phosphoglyceraldehyde to 3-phosphoglycerate

(2) 3-phosphoglyceraldehyde to

1,3-bisphosphoglycerate

(3) 3-phosphoglycerate to 2 phosphoglycerate

(4) None of these

83. What is the role of NAD^+ in cellular respiration :

(1) It functions as an enzymes

(2) It functions as an electron carrier

(3) It is a nucleotide source for ATP synthesis

(4) It is the final electron acceptor for anaerobic respiration

Title: Respiration in Plants

84. FAD acts as an electron acceptor in between :

(1) Fumaric and malic acid

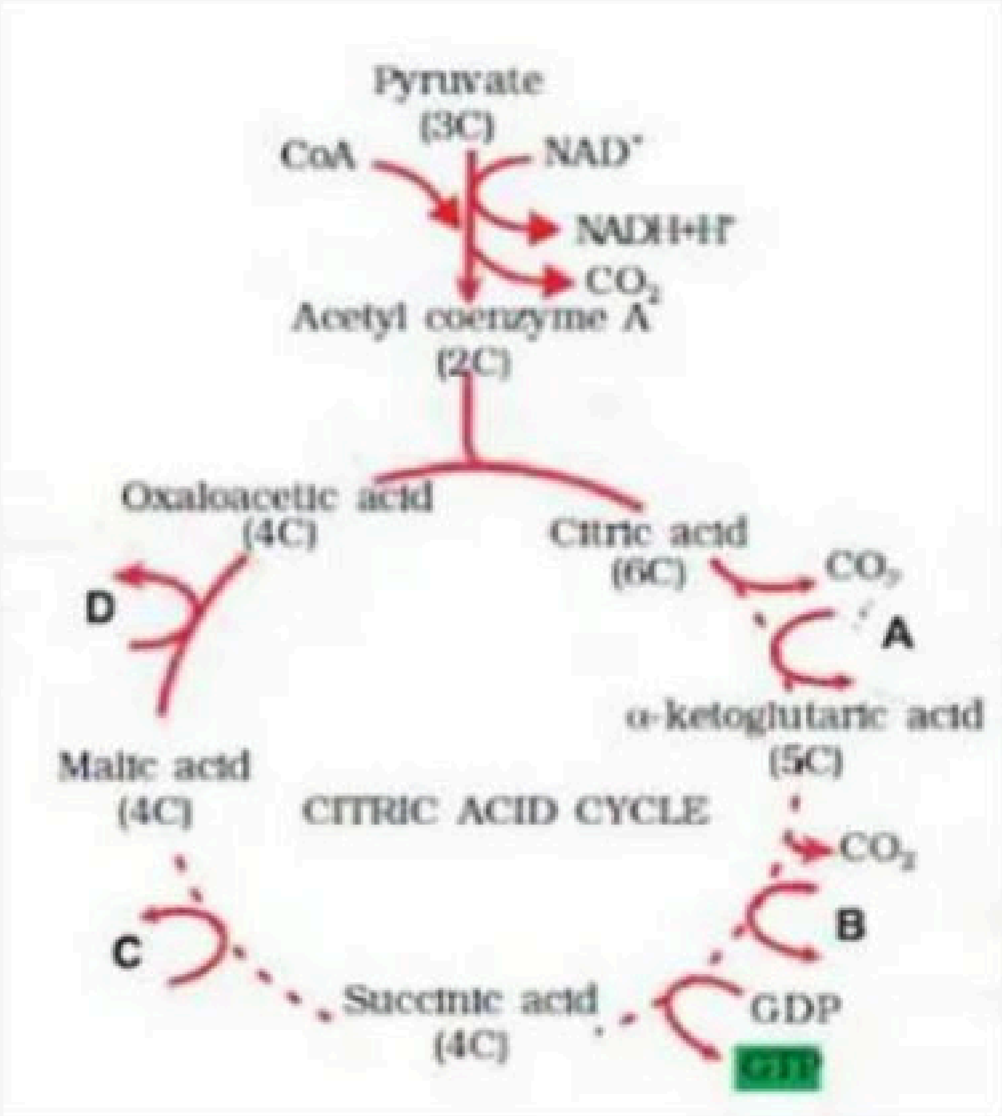
(2) Succinic and fumaric acid

(3) Malic and oxaloacetic acid

(4) Citric and isocitric acid

Title: Respiration in Plants

85. In given citric acid cycle which step/point FAD^{+} is reduced to $FADH_2$:



(1) A (2) B

(3) C (4) D

Title: Respiration in Plants

86. The crucial events in aerobic respiration are:

A. The complete oxidation of pyruvate by the stepwise removal of all the hydrogen atoms, leaving three molecules of CO_2.

B. The passing on of the electrons removed as part of the hydrogen atoms to molecular O_2 with simultaneous synthesis of ATP.

(1) First process is ETS and takes place in the matrix of the mitochondria, while the second process is TCA and takes place in the inner membrane of mitochondria.

(2) First process is TCA and takes place in the matrix of the mitochondria, while the second process is ETS and the inner membrane of mitochondria.

(3) First process is ETS and takes place in the inner membrane of mitochondria, while the second process is TCA and takes place in the matrix of the mitochondria.

(4) First process is TCA and takes place on the inner membrane of mitochondria, while the second process is ETS and takes place in the matrix of mitochondria.

Title: Respiration in Plants

87. Sequence of organic acids in Krebs' cycle is:

(1) Citric acid → oxalosuccinic acid → isocitric acid

(2) Citric acid → isocitric acid → oxalosuccinic acid

(3) Isocitric acid → oxalosuccinic acid → citric acid

(4) Oxalosuccinic acid → isocitric acid → citric acid

Title: Respiration in Plants

88. Find the values of w, x, y, and z in the given equation:

Pyruvic acid + wNAD + zFAD+ + XH_2O + ADP + iP → mitochondrial matrix → yCO_2 + wNADH + wH^+ + $zFADH_2$ + ATP

Options:

(1) w = 4, x = 3, y = 3, z = 1

(2) w = 3, x = 1, y = 2, z = 1

(3) w = 3, x = 2, y = 3, z = 2

(4) w = 4, x = 3, y = 3, z = 2

Title: Respiration in Plants

89. Out of 36 ATP molecules produced per glucose molecule during respiration:

(1) Two are produced outside glycolysis and 34 during the respiratory chain.

(2) Two are produced outside mitochondria and 34 inside mitochondria.

(3) Two are produced during glycolysis and 34 during the Krebs' cycle.

(4) All are formed inside mitochondria.

Title: Respiration in Plants

90. Protein is used as a respiratory substrate only when:

(1) Carbohydrates are absent.

(2) Fats are absent.

(3) Both carbohydrates and fats are absent.

(4) Fats and carbohydrates are abundant.

Title: Respiration in Plants

91. When malic acid is used as a respiratory substrate, the amount of CO_2 released is:

(1) More than O_2 consumed.

(2) Less than O_2 released.

(3) Equal to O_2 consumed.

(4) CO_2 is not released.

Title: Respiration in Plants

92. Match the columns and find out the correct combination:

Column A	Column B
A. Oxaloacetate	1. 6C-compound
B. Phosphoglyceraldehyde	2. 5C-compound
C. Isocitrate	3. 4C-compound
D. a-ketoglutarate	4. 3C-compound
	5. 2C-compound

Options:

(a) A-4, B-5, C-2, D-3

(b) A-3, B-4, C-1, D-2

(c) A-3, B-5, C-1, D-2

(d) A-2, B-4, C-1, D-5

Title: Respiration in Plants

93. Match the columns and find out the correct combination:

Column 1	Column 2
A. 4C-compound	1. Acetyl Co-A
B. 2C-compound	2. Pyruvate
C. 5C-compound	3. Citric acid
D. 3C-compound	4. a-Ketoglutaric acid
	5. Malic acid

Options:

(a) A-2, B-5, C-3, D-1

(b) A-3, B-1, C-4, D-2

(c) A-5, B-1, C-4, D-2

(d) A-5, B-3, C-1, D-2

Title: Respiration in Plants

94. Match the columns and find out the correct combination:

Column A	Column B
A. First formed stable intermediate of Krebs cycle	1. Succinyl CoA
B. Product of 2nd oxidation reaction	2. a-Ketoglutaric acid
C. Product of 1st decarboxylation reaction	3. Citric acid
D. Product of 2nd oxidative decarboxylation reaction	4. Isocitric acid
	5. Acetyl CoA

Options:

(a) A-3, B-4, C-2, D-1 (b) A-3, B-5, C-2, D-1

(c) A-3, B-5, C-4, D-2 (d) A-4, B-5, C-2, D-1

Title: Respiration in Plants

95. Match the following columns:

Column A	Column B
A. Molecular oxygen	1. α-ketoglutaric acid
B. Electron acceptor	2. Hydrogen acceptor
C. Pyruvate dehydrogenase	3. Cytochrome-c
D. Decarboxylation	4. Acetyl Co-A

Options:

(a) A-2, B-3, C-4, D-1

(b) A-3, B-4, C-2, D-1

(c) A-2, B-1, C-3, D-4

(d) A-4, B-3, C-1, D-2

Title: Respiration in Plants

96. Match the columns and find out the correct combination:

Electron Carrier	Number of Protons Carried per e Transfer	Gives e^- to
A. Ubiquinone	Two	Complex-III
B. Complex-II	Four	Ubiquinone
C. Complex-III	Two	Cytochrome c
D. Complex-V	ONE	Oxygen

Options:

(a) A and B (b) A and C

(c) C and D (d) B and D

Title: Respiration in Plants

97. Match the columns and find out the correct combination:

Stages of Respiration	Place of Occurrence	Products
A. Glycolysis	Cytosol	Pyruvic acid, $NADH_2$, ATP
B. Oxidative decarboxylation of pyruvic acid	Mitochondrial matrix	Acetyl CoA, $NADH_2$, CO_2
C. Krebs cycle	Peri-mitochondrial space	CO_2, $NADH_2$, ATP
D. Electron transport system	Inner membrane of mitochondria	ATP, CO_2

(a) A and B (b) C and D

(c) A and C (d) B and D

Title: Respiration in Plants

98. Select the wrong statement.

(a) When tripalmitin is used as a substrate in respiration, the RQ is 0.7.

(b) The intermediate compound which links glycolysis with the Krebs cycle is Acetyl CoA.

(c) One glucose molecule yields a net gain of 36 ATP molecules during fermentation.

(d) The scheme of glycolysis was given by Embden, Meyerhof, and Paranas.

99. Incorrect statement is:

(a) NADH is slowly oxidized to NAD^+ in fermentation.

(b) The F_1 headpiece in the peripheral membrane protein complex contains the site for the synthesis of ATP from ADP and inorganic phosphate.

(c) The number of ATP molecules synthesized in mitochondria during ETS depends on the nature of the electron donor.

(d) Less than 7% of the energy in glucose is released, and not all of it is trapped as ATP during fermentation.

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100. Which of the following statements regarding mitochondrial membrane is NOT correct?

(a) The inner membrane is highly convoluted, forming a series of infoldings.

(b) The outer membrane resembles a sieve.

(c) The outer membrane is permeable to all kinds of molecules.

(d) The enzymes of the electron transport chain are embedded in the outer mitochondrial membrane.

THANK YOU

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